

AP CALCULUS AB - UNIT 1

Guided Notes 1.10: Exploring Types of Discontinuities



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we classify removable, jump, and infinite discontinuities using one-sided evidence?

- Classify removable, jump, and infinite discontinuities using one-sided evidence.
- Distinguish the three named types from oscillatory or domain-gap failures.
- Use factor cancellation and multiplicity to classify rational-function discontinuities.
- Write the limit evidence before naming the type.

Key Knowledge, Vocabulary, and Notation

Removable	A finite common limit exists, but the function is missing there or has the wrong value.
Jump	Both one-sided limits are finite and unequal.
Infinite	At least one one-sided limit is $+\infty$ or $-\infty$.
Oscillatory	No single output destination exists and the behavior is not simply unbounded.
Classification protocol	Compute left limit, right limit, two-sided conclusion, then compare with the point value.

Concept Development and Strategy

1. For rational functions, factor first: complete cancellation creates a hole; a remaining denominator factor creates a pole.
2. For piecewise functions, evaluate the branch limits at every breakpoint.
3. Do not call every DNE limit a jump; identify the actual one-sided fingerprint.

Shared Representation / Data

Fingerprint	Classification
Finite common limit; point missing or wrong	Removable
Finite unequal one-sided limits	Jump
At least one unbounded side	Infinite
No finite/unbounded destination due to persistent oscillation	Other nonexistence

Worked Example: Removable rational discontinuity

Classify $f(x) = \frac{x^2-9}{x-3}$ at $x = 3$.

Answer and Reasoning

For $x \neq 3$, $f(x) = x + 3$, so the common finite limit is 6 while the original function is undefined. The discontinuity is removable.

Worked Example: Jump in a piecewise function

$g(x) = x + 1$ for $x < 2$ and $g(x) = x^2$ for $x \geq 2$. Classify at 2.

Answer and Reasoning

The left limit is 3 and the right limit is 4. They are finite and unequal, so the discontinuity is a jump.

Worked Example: Infinite by multiplicity

Classify $h(x) = \frac{x+1}{(x-4)^2}$ at $x = 4$.

Answer and Reasoning

The numerator is positive near 4 and the denominator is positive and tends to 0. Both sides tend to $+\infty$, so the discontinuity is infinite.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Classify $\frac{x^2-16}{x-4}$ at $x = 4$.

2. A graph has finite one-sided limits 2 and 5 at $x = 1$. Classify.

3. Classify every discontinuity of $f(x) = \frac{(x-3)(x+2)}{(x-3)(x-1)}$.

4. Classify $g(x) = \frac{x+2}{(x+2)^3}$ at $x = -2$.

5. Classify all discontinuities of $\frac{x^2-1}{(x-1)^2(x+3)}$.

6. Classify $f(x) = \frac{(x-a)^m}{(x-a)^n}$ at $x = a$ for positive integers m, n .

7. Construct one function having a removable discontinuity at -1 , a jump at 2 , and an infinite discontinuity at 4 .

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Which evidence identifies a jump discontinuity at $x = c$?

- (A) Both sides approach the same finite value. (B) Finite one-sided limits exist but are unequal.
(C) Both sides approach $+\infty$. (D) The function value is missing.

2. For $f(x) = \frac{(x-2)^2}{(x-2)^3}$, the discontinuity at 2 is

- (A) removable (B) jump
(C) infinite (D) none

3. A calculator graph of $\frac{x^2-4}{x^2-x-2}$ seems smooth except near $x = -1$ and $x = 2$. The correct classification is

- (A) holes at both points (B) poles at both points
(C) a pole at -1 and a hole at 2 (D) a hole at -1 and a pole at 2

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Prove that changing a function at a single point can repair only a removable discontinuity.

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